



INVESTIGATING SPATIAL STRUCTURES in Urban Environments FOR LEGAL AND POLICY INSIGHTS



Enhancing Brain Injury Surveillance Using Large Language Models and Topic Modeling

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The ISSUES Lab — Investigating Spatial Structures in Urban Environments



Who We Are

- A transdisciplinary research lab at The Ohio State University focused on how neighborhood and other environments affect health, safety, and justice.



What We Study

- Neighborhood-level risks and resilience
- Environmental exposures: disinvestment, surveillance, built environment
- Outcomes: child maltreatment, gun violence, IPV, adverse childhood experiences
- Data translation for public health, legal advocacy, and policy



Our Team

- Researchers and students from public health, law, social work, psychology, and geospatial science — united by a mission to strengthen families and promote equity.



Our Impact

- We produce actionable evidence to support lawyers, judges, practitioners, and policymakers in building healthier, safer communities.

Background



HEAD AND NECK TRAUMA ARE AMONG THE MOST COMMON AND SEVERE INJURIES AMONG SURVIVORS OF INTIMATE PARTNER VIOLENCE (IPV), RESULTING IN TRAUMATIC BRAIN INJURY (TBI)



EXISTING SURVEILLANCE SYSTEMS RARELY DISTINGUISH THE UNDERLYING MECHANISMS OF TBI IN IPV, LIMITING INSIGHT INTO CAUSAL MECHANISMS AND PREVENTION



THE NATIONAL ELECTRONIC INJURY SURVEILLANCE SYSTEM (NEISS) CAPTURES BOTH STRUCTURED INJURY CODES (E.G., BODY PART, DIAGNOSIS) AND UNSTRUCTURED FREE-TEXT NARRATIVES THAT CONTAIN RICH CONTEXTUAL DETAILS ABOUT INJURY CIRCUMSTANCES

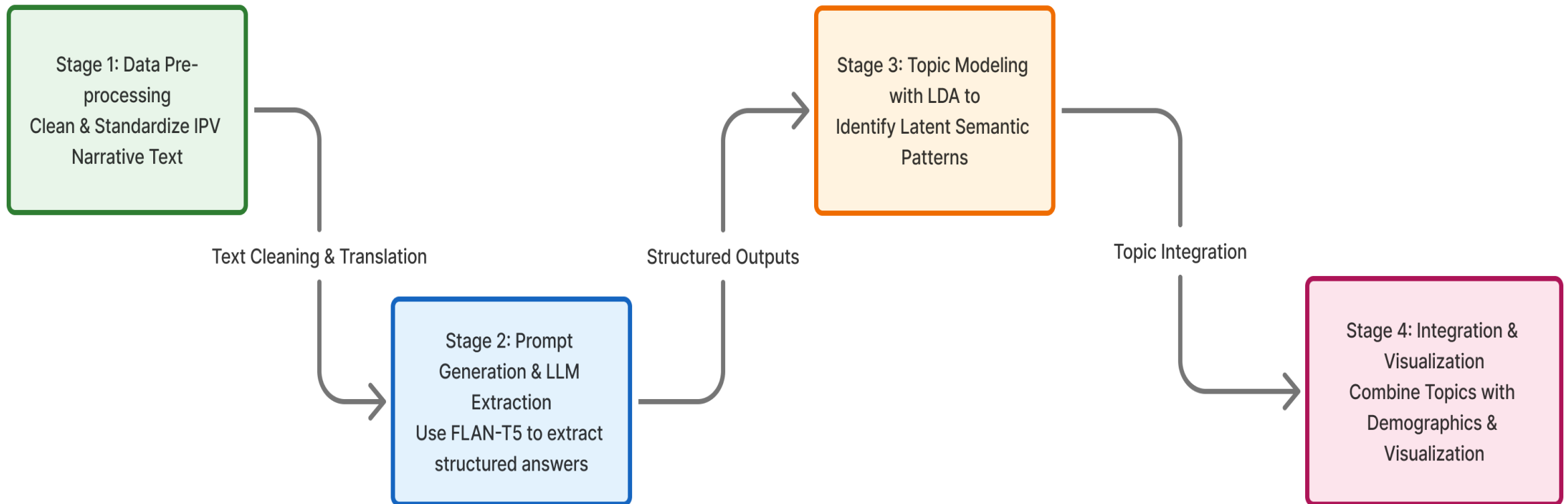


THE NEISS NARRATIVE FIELDS HAVE BEEN USED FOR <<SEE OUR PAPER>>, BUT THEY REMAIN UNDERUTILIZED FOR IDENTIFYING VIOLENCE-RELATED MECHANISMS OR TYPOLOGIES OF HARM

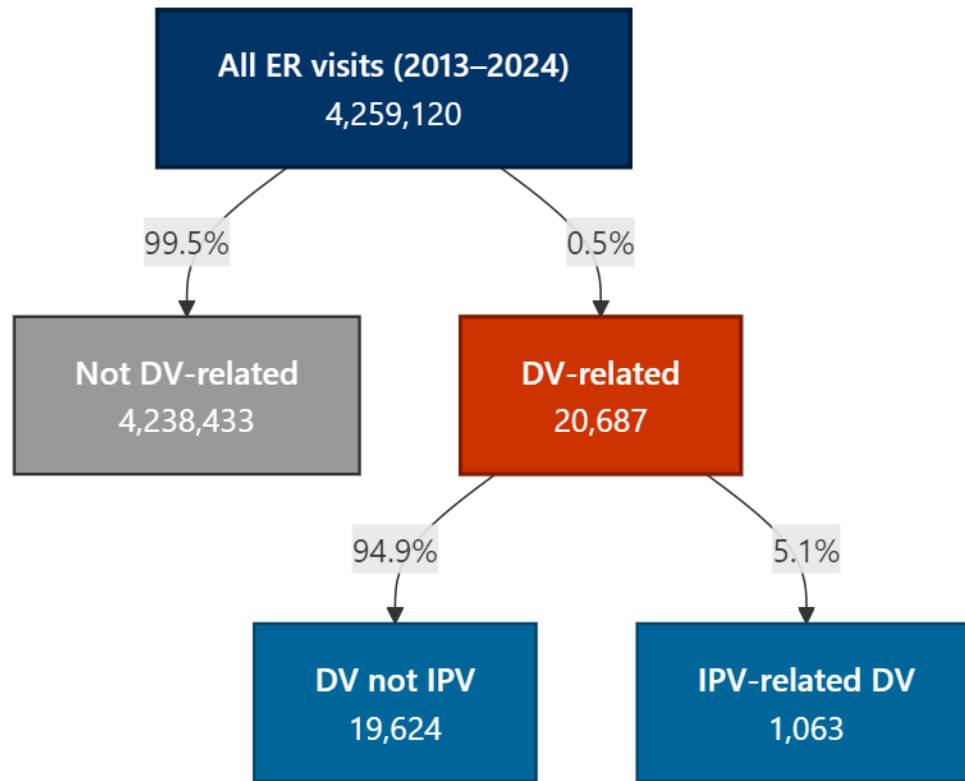
AI-Assisted Brain Injury Surveillance: Study Goals & Analytical Framework

Goal: To develop an artificial intelligence (AI) pipeline that applies a large language model (LLM) to extract and classify behavioral and contextual information from NEISS IPV narratives.

Aim: To identify and categorize head and neck injuries associated with IPV-related harm by integrating narrative-derived features with structured NEISS variables (e.g., diagnosis and injury location).



Identifying Intimate Partner Violence (IPV) in NEISS Narratives



NEISS narratives contain both explicit (e.g., “domestic violence”) and implicit indicators of conflict, aggression, or self-harm that are captured by the structured variables.

We created regular expression routines to flag narratives containing violence.

- We first used vectorized regex functions to remove common descriptors and expand abbreviations (e.g., lac = laceration) to transform the narratives to a human-readable format
- Within violence narratives, cases referencing relationship terms (e.g., *boyfriend*, *girlfriend*, *husband*, *wife*, *partner*) were identified as Intimate Partner Violence (IPV)
- IPV-related terms but excluding incidents involving intimates that were not IPV (e.g., accidental self-injury, agitation, and play fighting)

```
dict_abb_exp <- c(
  'pt'='patient', 'c/o'='complains of', 'lac'='laceration', 'rt'='right', 'l'='left', 'r'='right',
  's/p'='status post', 'w/'='with', '@'='at', 'fx'='fracture', 'loc'='loss of consciousness',
  'ed'='emergency department', 'sts'='states', 'acc'='accidentally', 'w'='with', 'hx'='history',
  'hd'='head', 'bib'='brought in by', 'lt'='left', 'inj'='injury', 'er'='emergency room',
  'bal'='blood alcohol level', 'fb'='foreign body', 'w/o'='without', 'pn'='pain',
  'nh'='nursing home', 'co'='complains of', 'p/w'='presents with', 'lwr'='lower',
  'str'='strained', 'spr'='sprain', 'sust'='sustained', 'h/o'='history of',
  's'd&f'='slipped, tripped, and fell', 't'd&f'='tripped and fell', 'bwd'='backward'
```

We generated custom prompts from the unstructured narratives by merging them with the structured variables to create human-readable text descriptions of the narratives (e.g., *patient, after argument w/ boyfriend, was locked out of house. tried to get into home by breaking glass window. +ethyl alcohol. clinical diagnosis: arm lacerations*).

Each prompt was then followed by a small set of targeted questions designed to capture key contextual and clinical details of each event.

```
# Cleaned narrative
narrative = str(row.get(TEXT_COL, "")) if pd.notna(row.get(TEXT_COL, "")) else ""

# Structured variables for context
age = row.get("age", "")
sex = str(row.get("sex", "")).lower()
diagnosis = str(row.get("diagnosis", "")).lower()
body_part = str(row.get("body_part", "")).lower()
disposition = str(row.get("disposition", "")).lower() if "disposition" in row else ""
location = str(row.get("location", "")).lower() if "location" in row else ""

# Expanded prompt
structured_context = (
    f"This record describes a {age}-year-old {sex} patient presenting to the emergency department. "
    f"The reported diagnosis was {diagnosis} affecting the {body_part}. "
    f"The patient was {disposition}. "
    f"The incident occurred at {location}. "
    f"Below is the clinician or patient description of the incident:\n\n"
    f"{narrative.strip()}"
)
```

LLM → LDA Output	Text
Raw NEISS Narrative	21YOF REPORTS SHE GOT INTO AN ARGUMENT W/ HER BOYFRIEND AND GRABBED A KITCHEN KNIFE AND PUT IT TO HER THROAT STATING SHE WAS GOING TO SLIT HER THROAT. PT STATES THIS WAS A MISTAKE AND SHE ACTED IMPULSIVELY IN THE MOMENT. CURRENTLY HAS NO SUICIDAL IDEATIONS OR SUICIDAL PLAN. CANNABINOID+, BAL <10. DX: SUICIDAL BEHAVIOR W/O ATTEMPTED SELF INJURY.
Cleaned Narrative Sent to LLM (Contextual prompt)	patient reports she got into an argument with her boyfriend and grabbed a kitchen knife, putting it to her throat while stating she was going to slit her throat. patient states this was a mistake and she acted impulsively in the moment. currently has no suicidal ideation or plan. cannabinoid positive, blood alcohol level less than 10. clinical diagnosis: suicidal behavior without attempted self-injury.\n\nQuestion: What was the person doing before the injury occurred?
LLM-Generated Structured Output	action: “arguing with boyfriend” cause: “grabbed kitchen knife and held it to throat” diagnosis: “suicidal behavior without self-injury” how: “impulsive self-harm gesture following argument” what: “kitchen knife” where: “home”

From LLM to LDA

We converted the unstructured free text into six targeted semantic representations of the same event.

- Each LDA model runs within one question column, discovering themes across all responses to that specific question
 - **Latent Cause** column → entries like *"wall and TV," "electric shock," "fall from stairs"*
 - **Interaction/Action** column → entries like *"arguing with boyfriend," "slipped and fell"*
 - **Medical outcomes/Diagnosis** column → entries like *"contusion," "fracture," "closed head injury"*
- Each LDA topic is converted into a short label and visualized across demographic groups to assess differences in topic prevalence

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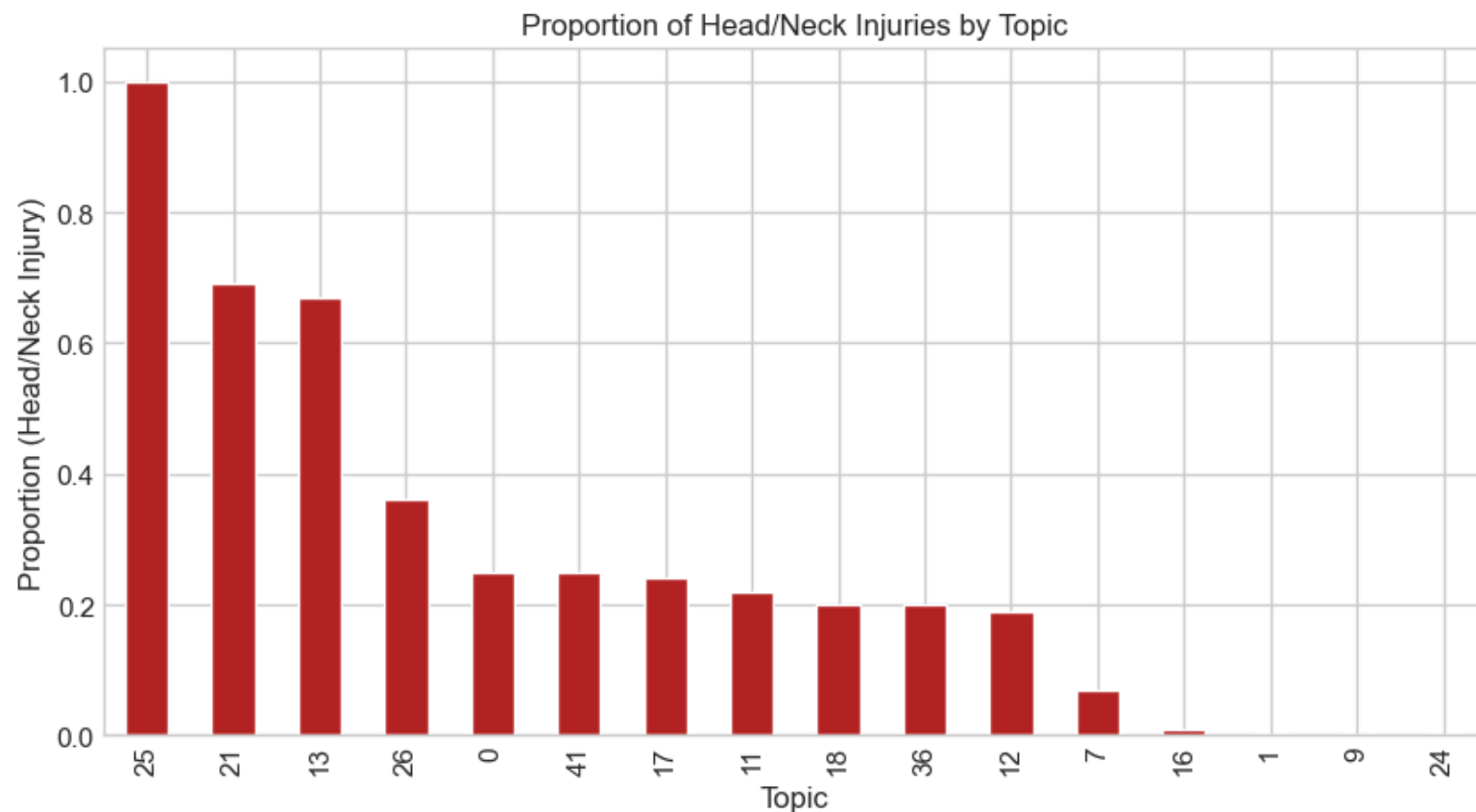
	action	cause	diagnosis	how	what	where
0	[arguing, with, boyfriend]	[hematoma]	[closed, head, injury]	[the, patient, was, arguing, with, boyfriend]	[slipped]	[carpet]
1	[in, an, argument, with, his, girlfriend]	[wall, and, tv]	[left, hand, contusion]	[the, patient, hit, wall, and, tv, with, fists]	[he, fell]	[wall]
2	[punching, window]	[electric, shock]	[hand, laceration]	[electric, shock]	[electric, shock]	[window]
3	[she, was, slammed, down, on, the, counter]	[shot, glass]	[the, patient, has, laceration, to, its, finger, cut]	[the, patient, slammed, shot, glass, down, on, the, counter, at, home, angry, with, husband]	[she, fell]	[on, the, counter]
4	[an, argument, with, boyfriend]	[an, argument, with, boyfriend]	[ear, ache]	[an, argument, with, boyfriend]	[he, shook, gas, can, at, her]	[on, its, foot]

Dimension	Top Words	Summary Interpretation
Action	argue, play, punch, verbal, wall, fight	Reflects interpersonal conflict and self-directed aggression during verbal altercations, often escalating to physical contact.
Cause	alcohol, intoxication, car, strike, window, glass	Indicates alcohol-related and blunt impact injuries, including falls and collisions linked to intoxication.
Diagnosis	hand, fractured, broken, injury, head, finger	Shows predominance of musculoskeletal and soft-tissue trauma, particularly fractures and contusions to the upper body.
How	behavior, play, fighting, punch, violent, interaction	Captures injuries arising from interpersonal struggle or impulsive acts during emotionally charged events.
What	wall, glass, window, brick, door, punch	Highlights impacts with household structures or objects commonly involved in physical altercations.
Where	home, outside, public, place, bathroom	Suggests most incidents occurred in the home or outside the home, with some in public or outdoor settings.

From LLM to LDA

Next, we identified which injury records are head/neck injuries and filtered the data to keep only those cases

This provided the prevalence of head/neck injury within each topic



[illegible][illegible]

A word cloud of injury-related terms. The word 'strike' is the largest and most prominent in the center. Other large words include 'fall', 'argue', 'crush', 'hit', 'fracture', 'hand', 'window', 'head', 'wall', 'bed', 'alcohol', 'abrasion', 'burn', 'door', 'glass', 'punch', 'car', 'fight', 'abuse', 'traumatic', 'slam', 'laceration', 'intoxication', 'alcoholic', 'splint', 'altercation', 'affect', 'hit', 'window', 'hand', 'fracture', 'alcohol', 'abrasion', 'burn', 'door', 'glass', 'punch', 'car', 'fight', 'abuse', 'traumatic', 'slam', 'laceration', 'intoxication', 'alcoholic', 'splint', 'altercation', 'affect'. The words are in various colors (green, yellow, orange, red, purple, blue) and orientations (horizontal, vertical, diagonal).

A word cloud of injury-related terms. The word 'head' is the largest and most prominent, centered in the image. Other large words include 'knee', 'close', 'injury', 'bone', 'finger', 'arm', 'leg', 'wrist', 'sprain', 'burn', 'traumatic', 'fractured', 'rib', 'laceration', 'syncope', 'broken', 'pain', 'thumb', 'break', 'forearm', 'amputation', 'avulsion', 'metacarpal', 'concussion', 'fall', 'leave', 'elbow', 'foot', 'low', 'contusion', 'concussion', 'bone', 'finger', 'arm', 'leg', 'wrist', 'sprain', 'burn', 'traumatic', 'fractured', 'rib', 'laceration', 'syncope', 'broken', 'pain', 'thumb', 'break', 'forearm', 'amputation', 'avulsion', 'metacarpal', 'concussion', 'fall', 'leave', 'elbow', 'foot', 'low', 'contusion', 'concussion'. The words are in various colors (blue, green, yellow, orange, red, purple) and orientations (horizontal, vertical, diagonal).

Injury Typology	%	Diagnosis Codes	Body Part Codes	Cause Topics	Representative Narrative
Accidental fall or slip	38.6%	52 (Concussion), 59 (Laceration), 62 (Closed-head injury)	75 (Head), 79 (Hip/Back), 76 (Face)	18, 21, 26	<i>"Patient was arguing with boyfriend, slipped and fell backward hitting head on carpet. Diagnosis: closed-head injury."</i>
Self-inflicted (intentional self-harm)	19.9%	53 (Contusion), 59 (Laceration), 62 (Head injury), 57 (Fracture)	75 (Head), 79 (Back), 76 (Face)	18, 21, 26, 13, 17	<i>"Patient began banging head against wall during argument with husband. Diagnosis: closed-head injury."</i>
Physical assault (partner-inflicted)	25.1%	59 (Laceration), 62 (Closed-head), 57 (Fracture), 71 (Pain)	75 (Head), 76 (Face), 79 (Abdomen)	18, 21, 12, 17	<i>"Patient was in argument with wife and she stabbed him in the neck with kitchen knife. Diagnosis: stab wound, neck."</i>
Blunt object impact	9.9%	59 (Laceration), 62 (Closed-head), 53 (Contusion), 52 (Concussion)	75 (Head), 76 (Face), 85 (Multiple sites)	18, 17, 21	<i>"Patient got angry at girlfriend and head-butted wall. Diagnosis: forehead abrasion, forehead hematoma."</i>
Environmental or chemical	2.3%	71 (Pain/irritation), 74 (Dermatitis)	76 (Face), 88 (Mouth), 94 (Ear)	18	<i>"Patient sprayed self with pepper spray while trying to spray boyfriend during fight. Diagnosis: contact dermatitis."</i>
Vehicle-related impact	1.7%	62 (Head injury), 57 (Fracture)	75 (Head), 79 (Back), 83 (Foot)	18, 36	<i>"Patient in argument with fiancé swerved and crashed into pole, complains of head and neck pain."</i>

Significance and Future Directions

Key Insights

- Combining regex typology (mechanism-based classification) and topic modeling (semantic clustering) enables a detailed mapping of *how, where, and why* brain injuries occur during an IPV-related incident.
- The AI-assisted pipeline transforms unstructured clinical narratives into quantitative data describing mechanisms of harm.

Public Health Relevance

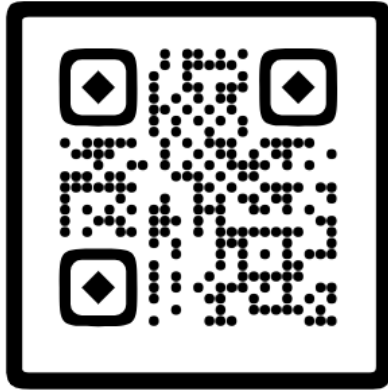
- Many “accidental” or “environmental” injuries co-occur with relationship conflict, revealing blurred boundaries between unintentional and interpersonal harm.
- Enhances injury surveillance and improves early detection of violence.
- Supports data-driven prevention and policy design targeting interpersonal and domestic injury risk.

Next Steps

- Validate typologies with larger datasets.
- Expand LLM prompts to capture more contextual cues.
- Examine patterns across sociodemographic and clinical subgroups.



<https://issues-osu.github.io/ISSUESlab/>



*Thank
you!*

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